Type MS · Finding r from sums of a geometric series

Answer 1 of 13 · Question 8 part a · 4 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

8 (a)	$a+ar=400$ or $\frac{a(1-r^2)}{1-r}=400,\quad ar+ar^2=100$ oe	B1B1
	Common methods 1) $r(a+ar)=100 \Rightarrow r(400)=100$	M1
	$r=\frac{1}{4}$*	A1cso
	2) $a=\frac{400}{1+r} \Rightarrow \left(\frac{400}{1+r}\right)r+\left(\frac{400}{1+r}\right)r^2=100$	M1
	$\Rightarrow 400r+400r^2=100+100r \Rightarrow 4r^2+3r-1=0 \Rightarrow (4r-1)(r+1)(=0)$ $r=\frac{1}{4}$*	A1cso
	3) $r=\frac{400-a}{a} \Rightarrow a\left(\frac{400-a}{a}\right)+a\left(\frac{400-a}{a}\right)^2=100 \Rightarrow$	M1
	$\Rightarrow 400-a+\frac{(400-a)^2}{a}=100 \Rightarrow$ $400-a^2+160000-800a+a^2=100a \Rightarrow a=320 \Rightarrow 320+320r=400$ $r=\frac{1}{4}$*	A1cso [4]
ALT	(Let G_1,G_2,G_3 be the first 3 terms) $G_2+G_3=100$	B1
	$(G_1+G_2=400) \Rightarrow rG_1+rG_2=100$	B1
	$rG_1+rG_2=100 \Rightarrow (r(G_1+G_2)=100) \Rightarrow r(400)=100$	M1
	$r=\frac{1}{4}$ *	A1*cso [4]
(b)	$(a=)\frac{300}{1-\left(\frac{1}{4}\right)^2}$ or $\frac{400}{1+\frac{1}{4}}$ or $\frac{100}{\frac{1}{4}+\left(\frac{1}{4}\right)^2}$ $=320$ *	M1A1 cso [2]
(c)	$S_\infty=\frac{320}{1-\frac{1}{4}}=\frac{1280}{3}$	M1A1 [2]
		Total 12 marks

Type MS · Finding r from sums of a geometric series

Answer 1 of 13 · Question 8 part a · 4 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For either equation shown correct a and r can be any letters throughout.
	B1	For both equations shown correct a and r can be any letters throughout.
	M1	For forming an equation eliminating a or r Allow one error in processing **such as a sign or arithmetical error, but not a 'cancellation'/simplification error**. Must be working with 2 correct equations. This mark can be awarded as soon as a or r are eliminated. Doesn't need simplification at this stage.
	A1 cso	For correctly solving and attaining $r = \frac{1}{4}$ minimum steps shown, no errors/omissions, ignore $r = -1$
ALT	B1	For either equation shown correct
	B1	For both equations shown correct
	M1	For multiplication of the first equation by r and formation of an equation in r Allow one error in processing. Must be working with 2 correct equations.
	A1	For $r = \frac{1}{4}$ minimum steps shown, no errors or omissions. Ignore work on any negative values
There are a number of different methods to do this, the four most commonly anticipated are shown. Mark to the following principles to gain the method mark: • One processing error only in any method (**M mark only, not A mark).** • Rearrange for r or a and correctly substitute into the other equation or such as method 1 to reach an equation in one variable only. • Rearrange the resulting equation so that an equation of the form $\boldsymbol{br = c}$ is reached. Note for the quadratic option, a factorisation will suffice. Note, if eliminating r, a must be found and the value of a then substituted into an appropriate equation. Methods where these principles can't be applied and thought worthy of credit – **send to review please**.		
(b)	M1	For using their expression for a with the correct r, to find a value for a Note, **for this question only**, work in (a) may be credited for this mark – **only if they eliminated r in their solution for part (a) and this is then used.**
	A1cso	For 320, no errors.
(c)	M1	For using the correct formula for the sum to infinity of a convergent series with the given values of a and r to find a value.
	A1	For the exact value of $\frac{1280}{3}$ oe or 426.67 or better (ie correctly rounded to more decimal places) or 426.6… (minimum 3 dots) or 426.66^{r} or $426.\dot{6}$
(d)	M1	Uses the correct formula for the sum of a geometric series, to set up **an inequality or equation,** allow < or > or = using the given values of r and a. Condone $\frac{1}{4}^{n}$
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left(\frac{1}{4}\right)^{n} < d \quad d \neq 0$ or $4^{n} < d$ Allow < or > or =. Dependent on the 1st method mark. Condone poor bracketing with powers again.
	ddM1	For the correct use of logs and correct use of an inequality sign throughout, including the reversal at the appropriate point. Dependent on both previous method marks. This mark may not be awarded if 'd' is negative. If candidates give a final answer of (n =) 7 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 7 Note although $n = 7$ can imply ddM1 as described, it is unlikely to imply the first 2 marks as there must be some logs work (directed by the question).

Type MS · Finding r from sums of a geometric series

Answer 1 of 13 · Question 8 part a · 4 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

(d)	$426.6 < \dfrac{320\left(1-\left(\frac{1}{4}\right)^n\right)}{1-\frac{1}{4}} \Rightarrow \left(\frac{1}{4}\right)^n < \frac{1}{6400}$ or $4^n < 6400$	M1dM1
	$\Rightarrow n > \dfrac{\log\left(\frac{1}{6400}\right)}{\log\left(\frac{1}{4}\right)}$ or $n > \log_{\frac{1}{4}}\left(\frac{1}{6400}\right)$ oe	ddM1
	$\Rightarrow n > 6.32... \Rightarrow n = 7$	A1 [4]
		Total 12 marks

Type MS · Finding r from sums of a geometric series

Answer 1 of 13 · Question 8 part a · 4 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 2 of 13 · Question 5 part a · 5 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Number		
5 (a)	$u_2 + u_4 = ar + ar^3 = 212.5$ $u_3 + u_4 = ar^2 + ar^3 = 62.5$	M1
	$\frac{(1+r^2)}{(r+r^2)} = \frac{17}{5}$	M1
	$12r^2 + 17r - 5 = 0$ $(4r-1)(3r+5) = 0$	M1 dM1
	$r = \frac{1}{4} \quad r = -\frac{5}{3}$	A1 (5)
(b)	$r = \frac{1}{4} \Rightarrow a = 800$ So $\frac{a}{1-r} = \frac{800}{\frac{3}{4}} = \frac{3200}{3}$	M1 A1 (2)
		Total 7 marks

Type MS · Finding r from sums of a geometric series

Answer 2 of 13 · Question 5 part a · 5 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For either $ar + ar^3 = 212.5$ or $ar^2 + ar^3 = 62.5$ correct
	M1	For attempting to eliminate a or ar either by division or substitution: e.g. $\frac{ar\left(1+r^2\right)}{ar\left(r+r^2\right)} = \frac{212.5}{62.5} \Rightarrow \frac{\left(1+r^2\right)}{\left(r+r^2\right)} = \frac{212.5}{62.5} = \left(\frac{17}{5}\right)$ An attempt involves some factorisation to eliminate a or ar
	Method 1 – finds a 3TQ	
	M1	For forming a 3TQ in r only using their expressions. $\left(12r^2 + 17r - 5 = 0 \text{ oe}\right)$ Accept for example $150r^2 + 212.5r - 62.5 = 0$
	dM1	For an attempt to solve their 3TQ to give **two** values of r See General Guidance for the definition of an attempt. For example: $(4r-1)(3r+5) = 0 \Rightarrow r = ..., ...$ This mark is dependent on the **FIRST** M mark being awarded
	Method 2 – finds a cubic equation	
	M1	For forming a cubic with a common factor of r in each term. e.g. $12r^3 + 17r^2 - 5r = 0$
	dM1	For factorising their cubic equation to achieve $r\left(12r^2 + 17r - 5\right) = 0$ and for an attempt to solve their 3TQ to give **two** values of r Ignore $r = 0$ if also given. See General Guidance for the definition of an attempt. For example: $(4r-1)(3r+5) = 0 \Rightarrow r = ..., ...$ This mark is dependent on the **FIRST** M mark being awarded
	A1	For the correct values; $r = \frac{1}{4}$ and $r = -\frac{5}{3}$ (reject $r = 0$ if seen earlier)
(b)	M1	Uses their $r = \frac{1}{4}$ [where $\lvert r \rvert < 1$] to find the value of a (800) with the correct formula for the sum of a geometric series to infinity. Condone an incorrect value of a even if they have used $r = \frac{1}{4}$ [The formula is given on page 2 of this booklet]. $S_\infty = \frac{a}{1-r} = \frac{'800'}{1-'\frac{1}{4}'} = ...$
	A1	For the correct value, $S_\infty = \frac{3200}{3}$ or $1066\frac{2}{3}$ Do not accept for example 1066.67 unless the stated value is $1066.\dot{6}$

Type MS · Finding r from sums of a geometric series

Answer 2 of 13 · Question 5 part a · 5 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 3 of 13 · Question 7 part b · 5 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

7(a)	$\frac{ar^7}{ar^6} = \frac{1152}{192} \quad (= 6) = r$	B1
	4th term $= \frac{192}{6^3}$ or $\frac{1152}{6^4} = \frac{8}{9}$	M1A1 (3)
(b)	$\frac{t_3}{r} + t_3 + rt_3 \implies \frac{24}{r} + 24 + 24r = -36$	M1A1 NB B1B1 on e-PEN
	$24 + 24r + 24r^2 = -36r$	dM1
	$24r^2 + 60r + 24 = 2r^2 + 5r + 2 = 0$ *	ddM1A1cso (5)
(c)	$2r^2 + 5r + 2 = 0 \Rightarrow (2r+1)(r+2) = 0 \Rightarrow r = -\frac{1}{2}$	M1A1
	$S = \frac{a}{1-r} = \frac{24 \div \left(-\frac{1}{2}\right)^2}{1-\left(-\frac{1}{2}\right)}, = 64$	M1,A1 (4)
		[12]

(a)	
B1	Obtain a correct value for r. Fraction need not be simplified
M1	Use their r and either the 7th or 8th term divided by the appropriate power of r to obtain the 4th term as a fraction – no need to simplify
A1	$\frac{8}{9}$
ALT	M1 Find a (=1/243) and use ar^3 A1 Correct answer
(b)	
M1	Use the given information to obtain an equation in r
A1	Correct equation
dM1	Eliminate the fraction
ddM1	Obtain a 3TQ, terms in any order
A1cso	Reach the given result with no errors in the working
(c)	
M1	Solve the given quadratic by any valid method. Must reach a value of r
A1	Correct value of r (Ignore second answer if given)
M1	Use the formula for the sum to infinity with their r **provided** $\lvert r \rvert < 1$. a must be 24 divided by (their r)^2
A1	Correct answer.

Type MS · Finding r from sums of a geometric series

Answer 3 of 13 · Question 7 part b · 5 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dy}{}$	

Type MS · Finding r from sums of a geometric series

Answer 4 of 13 · Question 8 part b · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

8 (a)	$a+ar=400$ or $\frac{a(1-r^2)}{1-r}=400,\quad ar+ar^2=100$ oe Common methods 1) $r(a+ar)=100 \Rightarrow r(400)=100$ $r=\frac{1}{4}$ *	B1B1 M1 A1cso
	2) $a=\frac{400}{1+r} \Rightarrow \left(\frac{400}{1+r}\right)r+\left(\frac{400}{1+r}\right)r^2=100$ $\Rightarrow 400r+400r^2=100+100r \Rightarrow 4r^2+3r-1=0 \Rightarrow (4r-1)(r+1)(=0)$ $r=\frac{1}{4}$ *	M1 A1cso
	3) $r=\frac{400-a}{a} \Rightarrow a\left(\frac{400-a}{a}\right)+a\left(\frac{400-a}{a}\right)^2=100 \Rightarrow$ $\Rightarrow 400-a+\frac{(400-a)^2}{a}=100 \Rightarrow$ $400-a^2+160000-800a+a^2=100a \Rightarrow a=320 \Rightarrow 320+320r=400$ $r=\frac{1}{4}$ *	M1 A1cso [4]
ALT	(Let G_1,G_2,G_3 be the first 3 terms) $G_2+G_3=100$ $(G_1+G_2=400) \Rightarrow rG_1+rG_2=100$ $rG_1+rG_2=100 \Rightarrow (r(G_1+G_2)=100) \Rightarrow r(400)=100$ $r=\frac{1}{4}$ *	B1 B1 M1 A1*cso [4]
(b)	$(a=)\frac{300}{1-\left(\frac{1}{4}\right)^2}$ or $\frac{400}{1+\frac{1}{4}}$ or $\frac{100}{\frac{1}{4}+\left(\frac{1}{4}\right)^2}$ $=320$ *	M1A1 cso [2]
(c)	$S_\infty=\frac{320}{1-\frac{1}{4}}=\frac{1280}{3}$	M1A1 [2]
		Total 12 marks

Type MS · Finding r from sums of a geometric series

Answer 4 of 13 · Question 8 part b · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For either equation shown correct a and r can be any letters throughout.
	B1	For both equations shown correct a and r can be any letters throughout.
	M1	For forming an equation eliminating a or r Allow one error in processing **such as a sign or arithmetical error, but not a 'cancellation'/simplification error**. Must be working with 2 correct equations. This mark can be awarded as soon as a or r are eliminated. Doesn't need simplification at this stage.
	A1 cso	For correctly solving and attaining $r = \frac{1}{4}$ minimum steps shown, no errors/omissions, ignore $r = -1$
ALT	B1	For either equation shown correct
	B1	For both equations shown correct
	M1	For multiplication of the first equation by r and formation of an equation in r Allow one error in processing. Must be working with 2 correct equations.
	A1	For $r = \frac{1}{4}$ minimum steps shown, no errors or omissions. Ignore work on any negative values
There are a number of different methods to do this, the four most commonly anticipated are shown. Mark to the following principles to gain the method mark: • One processing error only in any method (**M mark only, not A mark).** • Rearrange for r or a and correctly substitute into the other equation or such as method 1 to reach an equation in one variable only. • Rearrange the resulting equation so that an equation of the form $\boldsymbol{br = c}$ is reached. Note for the quadratic option, a factorisation will suffice. Note, if eliminating r, a must be found and the value of a then substituted into an appropriate equation. Methods where these principles can't be applied and thought worthy of credit – **send to review please**.		
(b)	M1	For using their expression for a with the correct r, to find a value for a Note, **for this question only**, work in (a) may be credited for this mark – **only if they eliminated r in their solution for part (a) and this is then used.**
	A1cso	For 320, no errors.
(c)	M1	For using the correct formula for the sum to infinity of a convergent series with the given values of a and r to find a value.
	A1	For the exact value of $\frac{1280}{3}$ oe or 426.67 or better (ie correctly rounded to more decimal places) or 426.6… (minimum 3 dots) or 426.66^{r} or $426.\dot{6}$
(d)	M1	Uses the correct formula for the sum of a geometric series, to set up **an inequality or equation,** allow < or > or = using the given values of r and a. Condone $\frac{1}{4}^{n}$
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left(\frac{1}{4}\right)^{n} < d \quad d \neq 0$ or $4^{n} < d$ Allow < or > or =. Dependent on the 1st method mark. Condone poor bracketing with powers again.
	ddM1	For the correct use of logs and correct use of an inequality sign throughout, including the reversal at the appropriate point. Dependent on both previous method marks. This mark may not be awarded if 'd' is negative. If candidates give a final answer of (n =) 7 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 7 Note although $n = 7$ can imply ddM1 as described, it is unlikely to imply the first 2 marks as there must be some logs work (directed by the question).

Type MS · Finding r from sums of a geometric series

Answer 4 of 13 · Question 8 part b · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

(d)	$426.6 < \dfrac{320\left(1-\left(\frac{1}{4}\right)^n\right)}{1-\frac{1}{4}} \Rightarrow \left(\frac{1}{4}\right)^n < \frac{1}{6400}$ or $4^n < 6400$	M1dM1
	$\Rightarrow n > \dfrac{\log\left(\frac{1}{6400}\right)}{\log\left(\frac{1}{4}\right)}$ or $n > \log_{\frac{1}{4}}\left(\frac{1}{6400}\right)$ oe	ddM1
	$\Rightarrow n > 6.32... \Rightarrow n = 7$	A1 [4]
		Total 12 marks

Type MS · Finding r from sums of a geometric series

Answer 4 of 13 · Question 8 part b · 2 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 5 of 13 · Question 8 part a · 7 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

8(a)	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe	M1M1 (B1B1on ePen
	$r = \left\{\frac{a(1+r)}{ar(1+r)} =\right\} \frac{288}{360} = \frac{4}{5}$	dM1A1
	$a = \frac{360}{1 + "\frac{4}{5}"} = 200$	M1A1
	$U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	A1cso(B1 on ePen) [7]
8(a) ALT	$a + ar = 360$ oe e.g. $\frac{a(1-r^2)}{1-r} = 360$ $ar + ar^2 = 288$ oe	M1M1 (B1B1 on ePen)
	$a\left(\frac{360}{a} - 1\right) + a\left(\frac{360}{a} - 1\right)^2 = 288$	dM1A1
	$\Rightarrow a = 200$	
	$r = \frac{360}{"200"} - 1 = \frac{4}{5}$	M1A1
	$U_n = 200\left(\frac{4}{5}\right)^{n-1}$ *	A1cso(B1 on ePen) [7]
(b)	{S is convergent because} $\left\|\frac{4}{5}\right\| < 1$	B1 [1]
(c)	$S_\infty = \frac{'200'}{1-'\left(\frac{4}{5}\right)'} = 1000$	M1A1cao [2]
(d)	$\frac{200\left(1-\left(\frac{4}{5}\right)^n\right)}{1-\left(\frac{4}{5}\right)} > 978$	M1
	$\left(\frac{4}{5}\right)^n < \frac{11}{500}$	dM1
	$n\,log\left(\frac{4}{5}\right) < log\left(\frac{11}{500}\right)$ $\Rightarrow n > 17.104\ldots$	ddM1
	$n = 18$	A1 [4]
		Total 14 marks

Type MS · Finding r from sums of a geometric series

Answer 5 of 13 · Question 8 part a · 7 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For either correct equation. Allow use of U_1 as the first term instead of a
	M1	For both correct equations. Allow use of U_1 as the first term instead of a
	dM1	For eliminating a from their equations, reaches r =... (Depends on previous mark)
	A1	For the correct value of r
	M1	Uses their value of r to find a value for a
	A1	For a = 200
	A1 cso	For the correct required expression of $\boldsymbol{U_n} = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(a)ALT	M1	For either correct equation.
	M1	For both correct equations
	dM1	For eliminating r from their equations, reaches a =... (Depends on previous mark)
	A1	For the correct value of a
	M1	Uses their value of a to find a value for r
	A1	For the correct value of r
	A1 cso	For the correct required expression of $U_n = 200\left(\frac{4}{5}\right)^{n-1}$ (must have $\boldsymbol{U_n}$)
(b)	B1	For stating the correct reason, $\lvert r\rvert < 1$ or -1 < r < 1 or "0.8" < 1 or $\left\lvert "\frac{4}{5}" \right\rvert < 1$ **Do not accept** just r < 1 without a correct reason
(c)	M1	For using their value of A /a /first term of their geometric sequence and their r (provided their $\lvert r\rvert < 1$)in a correct formula for the sum to infinity. $S_\infty = \frac{'their\ A'}{1-'(their\ r)'}$
	A1cao	For the correct value of 1000
(d)	M1	Uses their a and r (their r can be $\lvert r\rvert < 1$ or $\lvert r\rvert > 1$) to set the correct formula for the sum of a geometric series > 978 or =978 or ≥ 978 (>978 or =978 or ≥ 978 may be implied by later work.)
	dM1	Attempts to rearrange and achieves $(their\ r)^n$ " < "k (k *is non zero positive constant*) **Allow strict or non-strict inequalities or equation**: <,>, ≥ , ≤ or =
	ddM1	For correctly takes logs (any base)/ln, and achieves a positive value for n, it maybe implied by 17.1…but not 18 **ALT**: trial and error to find a value of n (at least tries n=17 and n=18)
	A1	For n = 18 Must come from **correct** working, achieves n=18 with no incorrect inequalities in their working.

If a candidate does not score the dM1 mark but arrives at an answer of 17.1...
this scores a maximum of M1 as clear use of calculator is not condoned.

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Answer 5 of 13 · Question 8 part a · 7 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 6 of 13 · Question 9 part a · 8 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

9(a)	$ar^2 = \frac{27}{2}$	B1
	$a + ar + \frac{27}{2} = \frac{57}{2}\left(\Rightarrow a + ar = 15 \Rightarrow a(1+r) = 15 \Rightarrow a = \frac{15}{1+r}\right)$	M1
	$\left(\frac{15}{1+r}\right)r^2 = \frac{27}{2} \Rightarrow 30r^2 - 27r - 27 = 0 \Rightarrow (10r^2 - 9r - 9 = 0)$	M1
	ALT $ar^2 = \frac{27}{2}$	B1
	$\frac{57}{2} = \frac{a(r^3-1)}{r-1}$	M1
	$\frac{57}{2} = \frac{27}{2r^2} \times \frac{(r^3-1)}{r-1} \Rightarrow 30r^3 - 57r^2 + 27 = 0$ $\left(\Rightarrow 10r^3 - 19r^2 + 9 = 0\right)$	M1
	r: $10r^2 - 9r - 9 = 0 \Rightarrow (5r+3)(2r-3) = 0 \Rightarrow r = \frac{3}{2}$	M1A1
	a: $a = \frac{15}{1+\frac{3}{2}} = 6$	A1
	$S_n = \sum_{r=1}^{n}\left('6' \div \frac{3}{2}\right)\left(\frac{3}{2}\right)^r \Rightarrow S_n = \sum_{r=1}^{n} 4\left(\frac{3}{2}\right)^r$ *	M1A1 cso [8]
(b)	$\frac{6\left(1.5^k - 1\right)}{1.5-1} > 50\,000$	M1
	$1.5^k > \frac{12\,503}{3} \Rightarrow \lg 1.5^k > \lg\frac{12\,503}{3}$	M1
	$k > \frac{\lg\frac{12\,503}{3}}{\lg\frac{3}{2}}$ or $k > \frac{\lg\frac{12\,503}{3}}{\lg 1.5}$ *	A1 [cso] [3]

Type MS · Finding r from sums of a geometric series

Answer 6 of 13 · Question 9 part a · 8 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

(c)	For $k > 20.556...$ so $k = 21$	B1 [1]
		Total 12 marks

Question	Scheme	Marks
9(a)	For stating, $ar^2 = \frac{27}{2}$	B1
	For summing the first three terms; $a + ar + \frac{27}{2} = \frac{57}{2}\left(\Rightarrow a + ar = 15 \Rightarrow a(1+r) = 15 \Rightarrow a = \frac{15}{1+r}\right)$ OR $\frac{57}{2} = \frac{a(r^3-1)}{r-1}$	M1
	For attempting to form a 3TQ using their two expressions for U_3 and S_3 $\left(\frac{15}{1+r}\right)r^2 = \frac{27}{2} \Rightarrow 30r^2 - 27r - 27 = 0 \Rightarrow (10r^2 - 9r - 9 = 0)$ OR For attempting to form a cubic using their two expressions for U_3 and S_3 $\frac{57}{2} = \frac{27}{2r^2} \times \frac{(r^3-1)}{r-1} \Rightarrow 30r^3 - 57r^2 + 27 = 0$	M1
	For attempting to solve their 3TQ (see general guidance). $10r^2 - 9r - 9 = 0 \Rightarrow (5r+3)(2r-3) = 0 \Rightarrow r = ...$ OR If following the alt method then this mark is awarded for factorising the cubic into a linear expression and a 3TQ **and** attempting to solve their 3TQ. $(r-1)(10r^2 - 9r - 9) = 0 \Rightarrow (r-1)(5r+3)(2r-3) = 0$ $\Rightarrow r = \cdots$	M1
	For the correct value of $r = \frac{3}{2}$	A1
	For the correct value of $a = \frac{15}{1+\frac{3}{2}} = 6$	A1
	For attempting to use their a and r to find the correct expression for S_n	

Type MS · Finding r from sums of a geometric series

Answer 6 of 13 · Question 9 part a · 8 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	$S_n = \sum_{r=1}^{n}\left('6' \div \frac{3}{2}\right)\left(\frac{3}{2}\right)^r \Rightarrow S_n = \ldots$ OR For demonstrating that a particular value of r gives the correct term in the sequence.	M1
	For the correct expression $S_n = \sum_{r=1}^{n} 4\left(\frac{3}{2}\right)^r$ * OR For demonstrating that a particular value of r gives the correct term in the sequence and commenting on a correct common ratio.	A1 cso [8]
(b)	For using the summation formula > 50 000 $\frac{6\left(1.5^k - 1\right)}{1.5 - 1} > 50\ 000$	M1
	For rearranging the inequality to achieve, $1.5^k > \frac{12\ 503}{3}$ and takes logs base 10 of both sides $\lg 1.5^k > \lg \frac{12\ 503}{3}$	M1
	For using the laws of logs to make k the subject. $k > \frac{\lg \frac{12\ 503}{3}}{\lg \frac{3}{2}}$ or $k > \frac{\lg \frac{12\ 503}{3}}{\lg 1.5}$ *	A1 cso [3]
(c)	For $k > 20.556\ldots \Rightarrow k = 21$	B1 [1]
		Total 12 marks

Type MS · Finding r from sums of a geometric series

Answer 6 of 13 · Question 9 part a · 8 marks

2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 7 of 13 · Question 7 part a.i · 0 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part a.i in this mark scheme extract

7 (a)	$\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ $\rightarrow$ **1**	M1
	$\frac{125}{6}=\frac{a}{1-r}$ $\rightarrow$ **2**	M1
(i)	$1\div 2\Rightarrow \frac{\frac{61}{6}}{\frac{125}{6}}=\frac{\frac{a\left(1-r^3\right)}{1-r}}{\frac{a}{1-r}}\Rightarrow \frac{61}{125}=1-r^3\Rightarrow r=\frac{4}{5}*$	dM1A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1
		[6]
	ALT	
	$a+ar+ar^2=a\ 1+r+r^2\ =\frac{61}{6}$ and $\frac{a}{1-r}=\frac{125}{6}$	[M1M1
	$\Rightarrow \frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61$	dM1
(i)	$\Rightarrow 1-r^3=\frac{61}{125}\Rightarrow r^3=\frac{64}{125}\Rightarrow r=\frac{4}{5}*$	A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1]
(b)	$19.8<\frac{\frac{25}{6}\left(1-0.8^n\right)}{1-0.8}\Rightarrow \frac{19.8\times 6\times 0.2}{25}<1-0.8^n$	M1
	$\frac{594}{625}<1-0.8^n\Rightarrow 0.8^n<\frac{31}{625}$	
	$n\lg(0.8)<\lg\left(\frac{31}{625}\right)*$	A1 cso
		[2]
(c)	$n>\frac{\lg\left(\frac{31}{625}\right)}{\lg(0.8)}\Rightarrow n>13.461...\Rightarrow n=14$	M1A1
		[2]
		Total 10 marks

Type MS · Finding r from sums of a geometric series

Answer 7 of 13 · Question 7 part a.i · 0 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part a.i in this mark scheme extract

	M1	For both $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. Eg., $\frac{125}{6}=\frac{a}{1-r}\Rightarrow a=\frac{125(1-r)}{6}\Rightarrow\frac{61}{6}=\frac{\left[\frac{125(1-r)}{6}\right]\left(1-r^3\right)}{1-r}\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1*	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	ALT	
(i)	**M1**	For either $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ or $\frac{125}{6}=\frac{a}{1-r}$ correct
	M1	For both $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. E.g. $\Rightarrow\frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	M1	For substituting $\frac{4}{5}$ into either **their** $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ or **their** $\frac{125}{6}=\frac{a}{1-r}$ or **their** $a+ar+ar^2=\frac{61}{6}$ (where their expression for a sum has been seen earlier) to find a value of a
	A1	For $a=\frac{25}{6}$
(b)	**M1**	For using the **correct** formula for the sum of a geometric series with $r=0.8$ [oe] and their a, setting up an inequality (allow < or > for this mark) using the value of 19.8 and attempting to achieve the given result.
	A1*	For the correct inequality as shown with no errors as this is a given result. $n\lg(0.8)<\lg\left(\frac{31}{625}\right)$
(c)	**M1**	For solving the given inequality in n using logarithms. They must achieve a value for n for this mark. Allow use of <, >, or = for this mark.
	A1	For $n=14$

Type MS · Finding r from sums of a geometric series

Answer 7 of 13 · Question 7 part a.i · 0 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 8 of 13 · Question 7 part a.ii · 6 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part a.ii in this mark scheme extract

7 (a)	$\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ $\rightarrow$ **1**	M1
	$\frac{125}{6}=\frac{a}{1-r}$ $\rightarrow$ **2**	M1
(i)	$1\div 2\Rightarrow \frac{\frac{61}{6}}{\frac{125}{6}}=\frac{\frac{a\left(1-r^3\right)}{1-r}}{\frac{a}{1-r}}\Rightarrow \frac{61}{125}=1-r^3\Rightarrow r=\frac{4}{5}*$	dM1A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1
		[6]
	ALT	
	$a+ar+ar^2=a\ 1+r+r^2\ =\frac{61}{6}$ and $\frac{a}{1-r}=\frac{125}{6}$	[M1M1
	$\Rightarrow \frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61$	dM1
(i)	$\Rightarrow 1-r^3=\frac{61}{125}\Rightarrow r^3=\frac{64}{125}\Rightarrow r=\frac{4}{5}*$	A1 cso
(ii)	$\frac{125}{6}=\frac{a}{1-\frac{4}{5}}\Rightarrow a=\frac{25}{6}$	M1A1]
(b)	$19.8<\frac{\frac{25}{6}\left(1-0.8^n\right)}{1-0.8}\Rightarrow \frac{19.8\times 6\times 0.2}{25}<1-0.8^n$	M1
	$\frac{594}{625}<1-0.8^n\Rightarrow 0.8^n<\frac{31}{625}$	
	$n\lg(0.8)<\lg\left(\frac{31}{625}\right)*$	A1 cso
		[2]
(c)	$n>\frac{\lg\left(\frac{31}{625}\right)}{\lg(0.8)}\Rightarrow n>13.461...\Rightarrow n=14$	M1A1
		[2]
		Total 10 marks

Type MS · Finding r from sums of a geometric series

Answer 8 of 13 · Question 7 part a.ii · 6 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part a.ii in this mark scheme extract

	M1	For both $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. Eg., $\frac{125}{6}=\frac{a}{1-r}\Rightarrow a=\frac{125(1-r)}{6}\Rightarrow\frac{61}{6}=\frac{\left[\frac{125(1-r)}{6}\right]\left(1-r^3\right)}{1-r}\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1*	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	ALT	
(i)	**M1**	For either $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ or $\frac{125}{6}=\frac{a}{1-r}$ correct
	M1	For both $a+ar+ar^2=\left[a\ 1+r+r^2\ \right]=\frac{61}{6}$ and $\frac{125}{6}=\frac{a}{1-r}$ correct
	dM1	For a valid method to eliminate a from both equations by division or substitution and attempting to re-arrange to find a value for r^3. E.g. $\Rightarrow\frac{125}{6}\ 1-r\ \ 1+r+r^2\ =\frac{61}{6}\Rightarrow 125\ 1-r^3\ =61\Rightarrow r^3=1-\frac{61}{125}$ This mark is dependent on **both** previous M marks.
	A1	For $r=\frac{4}{5}$ This value is given, there must be no errors for the award of this mark
	M1	For substituting $\frac{4}{5}$ into either **their** $\frac{61}{6}=\frac{a\left(1-r^3\right)}{1-r}$ or **their** $\frac{125}{6}=\frac{a}{1-r}$ or **their** $a+ar+ar^2=\frac{61}{6}$ (where their expression for a sum has been seen earlier) to find a value of a
	A1	For $a=\frac{25}{6}$
(b)	**M1**	For using the **correct** formula for the sum of a geometric series with $r=0.8$ [oe] and their a, setting up an inequality (allow < or > for this mark) using the value of 19.8 and attempting to achieve the given result.
	A1*	For the correct inequality as shown with no errors as this is a given result. $n\lg(0.8)<\lg\left(\frac{31}{625}\right)$
(c)	**M1**	For solving the given inequality in n using logarithms. They must achieve a value for n for this mark. Allow use of <, >, or = for this mark.
	A1	For $n=14$

Type MS · Finding r from sums of a geometric series

Answer 8 of 13 · Question 7 part a.ii · 6 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 9 of 13 · Question 7 part c · 3 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

7 (a)	$64\left(\frac{1}{4}\right)^3 - 64\left(\frac{1}{4}\right)^2 + 3 = 0$ so $(4x-1)$ is a factor	M1 A1 [2]
(b)	$16x^2 - 12x + "-3"$ $4x-1 \overline{)64x^3 - 64x^2 \{+0x\} + 3}$ $64x^3 - 16x^2$ $-48x^2$ or $64x^3 - 64x^2 + 3 = (4x-1)(16x^2 - 12x + C)$ $x = \frac{--12 \pm \sqrt{(-12)^2 - 4(16)(-3)}}{2\times 16}$ $x = \frac{1}{4}$ oe $x = \frac{3 \pm \sqrt{21}}{8}$ oe	M1 dM1 B1 A1 [4]
(c)	$ar^2 = 9$ or $\frac{a}{1-r} = 192$ oe $(192 - 192r)r^2 = 9$ oe $192r^2 - 192r^3 = 9 \Rightarrow 64r^3 - 64r^2 + 3 = 0$	B1 M1 A1*cso [3]
(d)	$r = \frac{1}{4}$	B1 [1]
(e)	$a\left(\frac{1}{4}\right)^2 = 9$ or $\frac{a}{1-\frac{1}{4}} = 192$ $\Rightarrow a = ...$ $a = 144$	M1 A1*cso [2]
(f)	$\frac{144\left(1-0.25^n\right)}{1-0.25} > 191.9$ $1 - 0.25^n > \frac{1919}{1920}$ $0.25^n < 0.00052083...$ $n > \frac{\log 0.00052083}{\log 0.25}$ $n > 5.45... \Rightarrow n = 6$	M1 dM1 M1 A1 [4]
		Total 16 marks

Type MS · Finding r from sums of a geometric series

Answer 9 of 13 · Question 7 part c · 3 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $64\left(\pm\frac{1}{4}\right)^3-64\left(\pm\frac{1}{4}\right)^2+3$
	A1	For = 0 and a conclusion stated, no errors seen. Accept minimal conclusion.
(b)	M1	Attempts long division. Minimally acceptable attempt is the division and correct working as written in the scheme. Must get a 3TQ factor. If comparing coefficients, a correct equation/comparison must be written followed by an attempt to find *C*. Must get a 3TQ factor.
	dM1	For a fully correct method to solve their 3TQ to find *x*, correct solutions from a correct 3TQ implies this mark.
	B1	For correct root $\frac{1}{4}$
	A1	For correct roots $\frac{3\pm\sqrt{21}}{8}$
(c)	B1	For either correct equation.
	M1	For both correct equations and any correct step to eliminate a
	A1*cso	For a fully correct solution, no errors, with at least one correct intermediate line leading to $64r^3-64r^2+3=0$
(d)	B1 **(A1 on ePen)**	For the correct value of r
(e)	M1	For correct substitution of their value for r in either of their equations to find a
	A1	For $a = 144$
(f)	M1	**Allow working in equation or any inequality for all M marks** Correct use of sum to n terms formula with $a = 144$ or their a and their r to set up an equation or inequality
	dM1	Attempts to rearrange and obtains $\pm(\text{"their } r\text{"})^n \ldots p$
	M1	Correct use of logs and attempt to evaluate the log. (Can be implied by awrt 5.5 only)
	A1	$n = 6$ (must come from solving inequalities with no errors in processing the inequalities. If from solving equations only, award A0)

Type MS · Finding r from sums of a geometric series

Answer 9 of 13 · Question 7 part c · 3 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 10 of 13 · Question 7 part d · 1 mark

2024 NOV · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

7 (a)	$64\left(\frac{1}{4}\right)^3 - 64\left(\frac{1}{4}\right)^2 + 3 = 0$ so $(4x-1)$ is a factor	M1 A1 [2]
(b)	$16x^2 - 12x + "-3"$ $4x-1 \overline{)64x^3 - 64x^2\{+0x\} + 3}$ $64x^3 - 16x^2$ $-48x^2$ or $64x^3 - 64x^2 + 3 = (4x-1)(16x^2 - 12x + C)$ $x = \frac{--12 \pm \sqrt{(-12)^2 - 4(16)(-3)}}{2\times 16}$ $x = \frac{1}{4}$ oe $x = \frac{3 \pm \sqrt{21}}{8}$ oe	M1 dM1 B1 A1 [4]
(c)	$ar^2 = 9$ or $\frac{a}{1-r} = 192$ oe $(192 - 192r)r^2 = 9$ oe $192r^2 - 192r^3 = 9 \Rightarrow 64r^3 - 64r^2 + 3 = 0$	B1 M1 A1*cso [3]
(d)	$r = \frac{1}{4}$	B1 [1]
(e)	$a\left(\frac{1}{4}\right)^2 = 9$ or $\frac{a}{1-\frac{1}{4}} = 192$ $\Rightarrow a = ...$ $a = 144$	M1 A1*cso [2]
(f)	$\frac{144\left(1-0.25^n\right)}{1-0.25} > 191.9$ $1 - 0.25^n > \frac{1919}{1920}$ $0.25^n < 0.00052083...$ $n > \frac{\log 0.00052083}{\log 0.25}$ $n > 5.45... \Rightarrow n = 6$	M1 dM1 M1 A1 [4]
		Total 16 marks

Type MS · Finding r from sums of a geometric series

Answer 10 of 13 · Question 7 part d · 1 mark

2024 NOV · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $64\left(\pm\frac{1}{4}\right)^3-64\left(\pm\frac{1}{4}\right)^2+3$
	A1	For = 0 and a conclusion stated, no errors seen. Accept minimal conclusion.
(b)	M1	Attempts long division. Minimally acceptable attempt is the division and correct working as written in the scheme. Must get a 3TQ factor. If comparing coefficients, a correct equation/comparison must be written followed by an attempt to find *C*. Must get a 3TQ factor.
	dM1	For a fully correct method to solve their 3TQ to find *x*, correct solutions from a correct 3TQ implies this mark.
	B1	For correct root $\frac{1}{4}$
	A1	For correct roots $\frac{3\pm\sqrt{21}}{8}$
(c)	B1	For either correct equation.
	M1	For both correct equations and any correct step to eliminate a
	A1*cso	For a fully correct solution, no errors, with at least one correct intermediate line leading to $64r^3-64r^2+3=0$
(d)	B1 **(A1 on ePen)**	For the correct value of r
(e)	M1	For correct substitution of their value for r in either of their equations to find a
	A1	For $a = 144$
(f)	M1	**Allow working in equation or any inequality for all M marks** Correct use of sum to n terms formula with $a = 144$ or their a and their r to set up an equation or inequality
	dM1	Attempts to rearrange and obtains $\pm(\text{"their } r\text{"})^n \ldots p$
	M1	Correct use of logs and attempt to evaluate the log. (Can be implied by awrt 5.5 only)
	A1	$n = 6$ (must come from solving inequalities with no errors in processing the inequalities. If from solving equations only, award A0)

Type MS · Finding r from sums of a geometric series

Answer 10 of 13 · Question 7 part d · 1 mark

2024 NOV · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 11 of 13 · Question 7 part e · 2 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part e in this mark scheme extract

7 (a)	$64\left(\frac{1}{4}\right)^3 - 64\left(\frac{1}{4}\right)^2 + 3 = 0$ so $(4x-1)$ is a factor	M1 A1 [2]
(b)	$16x^2 - 12x + "-3"$ $4x-1\overline{)64x^3 - 64x^2\{+0x\} + 3}$ $64x^3 - 16x^2$ $-48x^2$ or $64x^3 - 64x^2 + 3 = (4x-1)(16x^2 - 12x + C)$	M1
	$x = \frac{--12 \pm \sqrt{(-12)^2 - 4(16)(-3)}}{2\times 16}$	dM1
	$x = \frac{1}{4}$ oe	B1
	$x = \frac{3 \pm \sqrt{21}}{8}$ oe	A1 [4]
(c)	$ar^2 = 9$ or $\frac{a}{1-r} = 192$ oe	B1
	$(192 - 192r)r^2 = 9$ oe	M1
	$192r^2 - 192r^3 = 9 \Rightarrow 64r^3 - 64r^2 + 3 = 0$	A1*cso [3]
(d)	$r = \frac{1}{4}$	B1 [1]
(e)	$a\left(\frac{1}{4}\right)^2 = 9$ or $\frac{a}{1-\frac{1}{4}} = 192$ $\Rightarrow a = ...$	M1
	$a = 144$	A1*cso [2]
(f)	$\frac{144\left(1-0.25^n\right)}{1-0.25} > 191.9$	M1
	$1 - 0.25^n > \frac{1919}{1920}$	
	$0.25^n < 0.00052083...$	dM1
	$n > \frac{\log 0.00052083}{\log 0.25}$	M1
	$n > 5.45... \Rightarrow n = 6$	A1 [4]
		Total 16 marks

Type MS · Finding r from sums of a geometric series

Answer 11 of 13 · Question 7 part e · 2 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part e in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $64\left(\pm\frac{1}{4}\right)^3-64\left(\pm\frac{1}{4}\right)^2+3$
	A1	For = 0 and a conclusion stated, no errors seen. Accept minimal conclusion.
(b)	M1	Attempts long division. Minimally acceptable attempt is the division and correct working as written in the scheme. Must get a 3TQ factor. If comparing coefficients, a correct equation/comparison must be written followed by an attempt to find *C*. Must get a 3TQ factor.
	dM1	For a fully correct method to solve their 3TQ to find *x*, correct solutions from a correct 3TQ implies this mark.
	B1	For correct root $\frac{1}{4}$
	A1	For correct roots $\frac{3\pm\sqrt{21}}{8}$
(c)	B1	For either correct equation.
	M1	For both correct equations and any correct step to eliminate a
	A1*cso	For a fully correct solution, no errors, with at least one correct intermediate line leading to $64r^3-64r^2+3=0$
(d)	B1 **(A1 on ePen)**	For the correct value of r
(e)	M1	For correct substitution of their value for r in either of their equations to find a
	A1	For $a = 144$
(f)	M1	**Allow working in equation or any inequality for all M marks** Correct use of sum to n terms formula with $a = 144$ or their a and their r to set up an equation or inequality
	dM1	Attempts to rearrange and obtains $\pm(\text{"their } r\text{"})^n \ldots p$
	M1	Correct use of logs and attempt to evaluate the log. (Can be implied by awrt 5.5 only)
	A1	$n = 6$ (must come from solving inequalities with no errors in processing the inequalities. If from solving equations only, award A0)

Type MS · Finding r from sums of a geometric series

Answer 11 of 13 · Question 7 part e · 2 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part e in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding r from sums of a geometric series

Answer 12 of 13 · Question 4 part whole · 7 marks

2022 JUN · Paper 01 · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding r from sums of a geometric series

Answer 13 of 13 · Question 7 part a · 2 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

7(a)	$\left(\frac{ar^2}{a}=r^2=\right)\frac{2704}{625}$ oe	M1
	$(r=)\pm\frac{13}{25}$ oe	A1 [2]
(b)	$(S_\infty=)\frac{16}{1-"\frac{13}{25}"}=\frac{100}{3}$ oe	M1A1 [2]
(c)	$\frac{16\left(1-\left("\frac{13}{25}"\right)^n\right)}{1-"\frac{13}{25}"}>33\left(\Rightarrow 1-\left(\frac{13}{25}\right)^n>0.99\right)$ oe	M1
	$\Rightarrow\left("\frac{13}{25}"\right)^n<0.01$ oe	dM1
	$n\lg\left("\frac{13}{25}"\right)<\lg(0.01)$ oe	M1
	$\Rightarrow n>\frac{\lg(0.01)}{\lg\left("\frac{13}{25}"\right)}\Rightarrow n>7.04...\text{oe}\Rightarrow(n=)8$	dddM1A1 [5]
		Total 9 marks

Type MS · Finding r from sums of a geometric series

Answer 13 of 13 · Question 7 part a · 2 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For any correct unsimplified expression for r or r^2.
	A1	For $r = \pm\frac{13}{25}$ oe
(b)	M1	Uses the correct formula for the sum to infinity of a geometric series using their positive $r < 1$ and the given first term.
	A1	For $\frac{100}{3}$ oe . Allow 33.33……, 33.3^r, $33.\dot{3}$ etc. The question demand is for an exact answer so there must be some indication of recurrence. This could be min 1 dp with dots after ie 33.3….., but cannot be a terminating decimal.
(c)	M1	Uses the correct formula for the sum of a geometric series to set up an **inequality or equation** in terms of n, *ft* their **positive** $r < 1$, must be using $a = 16$ Allow < or > or =
	dM1	For simplifying (allow errors in simplification) their **inequality or equation** in n to the form $\left("\frac{13}{25}"\right)^n < d \quad d \neq 0$. Allow $"\frac{13}{25}"$ to be their positive $r < 1$. Allow < or > or =. Dependent on the 1st method mark.
	M1	Takes logarithms (any base) of their **exponential equation or inequality <u>correctly</u>** on both sides and correctly uses the power law to reach $n\lg(a) < \lg(b)$. or 'de-logs' their **exponential equation or inequality <u>correctly</u>** to get $\log_e d < e$ This isn't a dependent mark, but the candidate must be able to correctly take logs of both sides and use the power law or correctly de-log their equation/inequality. r can be any value carried through from part (a) for this mark. Allow < or > or =, the inequality doesn't need to have been reversed at this point.
	dddM1	For finding a value for n setting up the correct inequality (with their positive $r < 1$) from the beginning and where the inequality sign has been reversed at the appropriate point during their work. Dependent on all previous method marks. If candidates give a final answer of (n =) 8 – this mark can be implied even if the inequality sign is not correctly reversed.
	A1	For (n =) 8

Type MS · Finding r from sums of a geometric series

Answer 13 of 13 · Question 7 part a · 2 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks